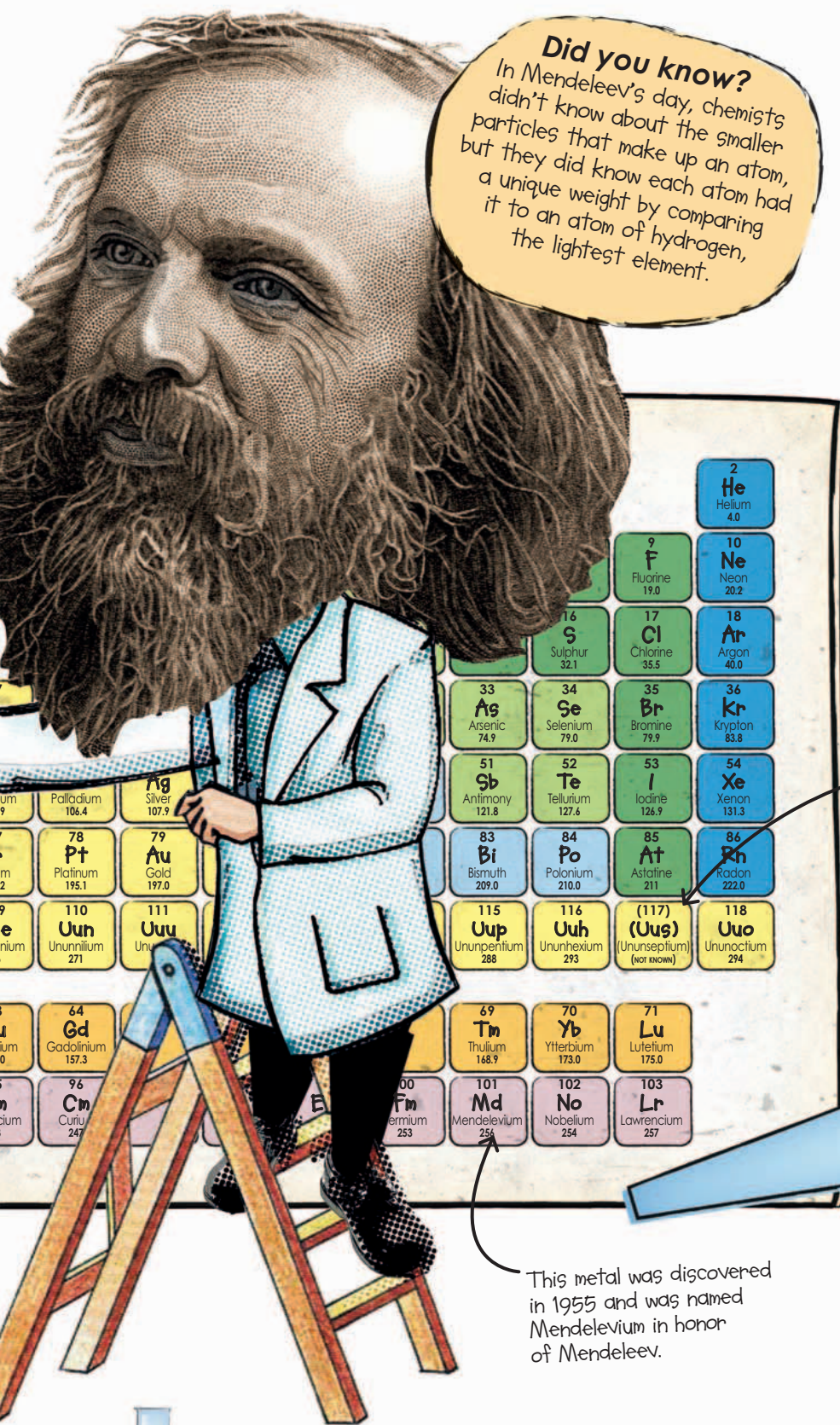




Did you know?

In Mendeleev's day, chemists didn't know about the smaller particles that make up an atom, but they did know each atom had a unique weight by comparing it to an atom of hydrogen, the lightest element.

Order out of chaos

Mendeleev believed there must be some kind of order to the elements. Previously, chemists had grouped the elements either by their atomic weight or by how they behaved (what they reacted with). Mendeleev turned it into a game of cards that he called "**chemical solitaire**." He wrote each element's name and atomic weight on a card and tried to organize them. Eventually, he had a table that listed the elements by their atomic weight and grouped them into **nine families**, such as metals, nonmetals, and noble gases. The periodic table he created in 1869 revealed that there was a **PATTERN** to the way the elements behaved.

Mendeleev left gaps for elements that were **STILL UNKNOWN**. By looking at the patterns of his table, he predicted the appearance and behavior of the missing elements. He was right every time.

A science revolution

The periodic table of elements completely revolutionized our understanding of the properties of atoms and led to the discovery of many elements new to science. It also paved the way for a new branch of science called quantum mechanics.

This metal was discovered in 1955 and was named Mendelevium in honor of Mendeleev.

The first ever list of the chemical elements was put together by the French chemist **ANTOINE LAVOISIER** (1743–1794).

The Italian chemist **STANISLAO CANNIZZARO** (1826–1910) devised a way to measure the atomic weights of the elements.

